

1a.

Clustering is an unsupervised machine learning technique used to group similar data points together based on shared feature characteristics. It differs from supervised learning in that it doesn't rely on labeled data. Instead, it uses patterns or structures within the dataset itself. This section touched on clustering's ability to uncover hidden groupings within data, which is useful for exploratory data analysis, segmentation, and understanding relationships between various observations at hand.

1b.

Clustering algorithms consist of different methods used to group data points. They each contain their own approach and come with their own assumptions that dictate things like centroid-based methods (k-means), density-based methods (DBSCAN), and hierarchical clustering. No single algorithm works best for all datasets, and there may be situations where one works better than the other, depending on the structure of the data and desired output for the problem being observed.

1c.

The clustering workflow outlines the typical steps followed when applying clustering to a dataset. The steps included defining the problem, preparing the data, selecting a similarity measure, choosing an algorithm, evaluating results, and refining the model accordingly. Clustering is an iterative process, and results often improve through repeated adjustments and experimentation with data and processing.

1d.

Data preparation involves cleaning, transforming, and organizing data before applying clustering algorithms. The handling of missing values, normalizing and standardized scaling, removal of outliers, and more all go into this. Clustering can easily be highly sensitive to imbalanced data of poor quality and improper scale, so proper preprocessing is necessary to produce meaningful and accurate clusters.

1e.

K-means clustering is a popular centroid-based algorithm that partitions data into k distinct clusters by minimizing the distance between data points and their assigned cluster centers. It works iteratively by assigning points to the nearest centroid and updating centroids based on the average of assigned points. K-means is efficient and easy to implement, but it requires specifying the number of clusters in advance.

1f.

A manual similarity measure refers to defining how similar or different two data points are, often using distance metrics like Euclidean distance. This measure determines how clusters are formed. The choice of similarity metric directly affects clustering results, so it must align with the nature of the data and the problem.

1g.

Evaluating clustering results involves assessing how well the algorithm grouped the data. Since clustering is unsupervised, evaluation can be challenging, but techniques like the elbow method, silhouette score, and visual inspection are commonly used. The evaluation is subjective and often requires combining multiple methods to determine cluster quality.

1h.

Clustering has several advantages, such as discovering hidden patterns, working without labeled data, and being useful for exploratory analysis. However, it also has disadvantages, including sensitivity to noise, difficulty in choosing parameters (like the number of clusters), and challenges in evaluation. While clustering is powerful, it requires careful application and interpretation.

2.

These sections of the course focus on introducing clustering and explaining how k-means works in practice. The goal is to help learners understand how to group data, choose the right approach, and evaluate the results. Overall, the course is meant to build a basic understanding of clustering and how it's used in real-world data analysis.

3.

This course helped me better understand the practical side of clustering. In lectures, we talk about the concepts, but this went more into how to actually apply them, like choosing similarity measures and evaluating results. It also made it clearer that clustering involves a lot of trial and error (iterations).

4.

The material was organized in a way that was easy to follow, and the explanations were straightforward. I also liked that it broke things into smaller sections instead of overwhelming everything at once. This made it easier to stay focused and understand each concept.

5.

I think the course could be improved by adding more real-world examples or showing a full example from start to finish. While the explanations were clear, seeing everything applied in one complete scenario would make it easier to connect all the steps together.